

4. Write a procedure in MIPS code that takes the base address of a string as an input parameter and counts and return the number of 'e's found in the string (for example, the string "engine" has 2 'e's'). (15 marks)

Hint : Assume that the ASCII format is used for storing the string. The ASCII value of e is 101.

Solution:

Show all steps taken to obtain the answer. Absence of steps in your answer will result in the absence of marks for your answer.

- i) What number does the following set of bits (specified in the IEEE 754) represent
 11111111100111110101101011110000

single

$$127 + X = 255$$

$$X = 128$$

$$1.0011111010110101110000 \times 2^{128}$$

- ii) Represent 1024.00 in the IEEE 754 double-precision format

$$1.0000000000 \rightarrow 1.0000000000$$

2. Translate the following statement in C/C++ to binary code (into 0s and 1s) (SO-1, 10 marks)

$$A[14] = (x/13 - 7*B[12]);$$

Assume that variable x is stored in register \$t3, and base addresses of A and B are stored in \$s4 and \$s5, respectively.

Note: Use the information provided at the end of the paper where needed.

1. Circle the correct answer in the following MCQs

a) All are examples of SaaS, except

- i) Blackboard
- ii) Google Docs
- iii) Google Chrome
- iv) ODUS

b) _____ is used to make a system more dependable

- i) Redundancy
- ii) Parallelism/Pipelining
- iii) Abstraction
- iv) Memory hierarchy

c) _____ helped increase programmer productivity.

- i) SaaS
- ii) Parallelism
- iii) Advent of high-level languages
- iv) Machine code

d) A wafer manufactured using 16nm technology used to manufacture 320 dies had a yield. What is the number of damaged dies in this case?

- i) $(320) * (97.5/100) * 0.16$
- ii) 8
- iii) $(16 * 10^{-9} * 320) * (97.5/100)$
- iv) 312

e) Moore's Law states that

- i) the IC resources double every 18-24 months
- ii) the processor yield increases every 6-12 months
- iii) new manufacturing cycle always gives a better yield
- iv) it is important to make the common case fast

5. Explain the use and benefit of PC-relative addressing in MIPS with a suitable example.
(10 marks)

... number to the specified representation